

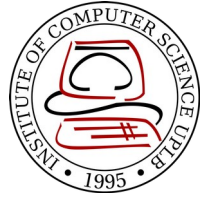
CMSC 137

Data Communications and Networking

Lecture Module
16 – Wired LAN: Ethernet



University of the Philippines
LOS BAÑOS



Module 16: Wired LAN: Ethernet

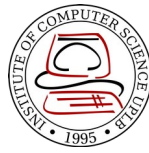
Rozano S. Maniaol
Lecturer

Introduction



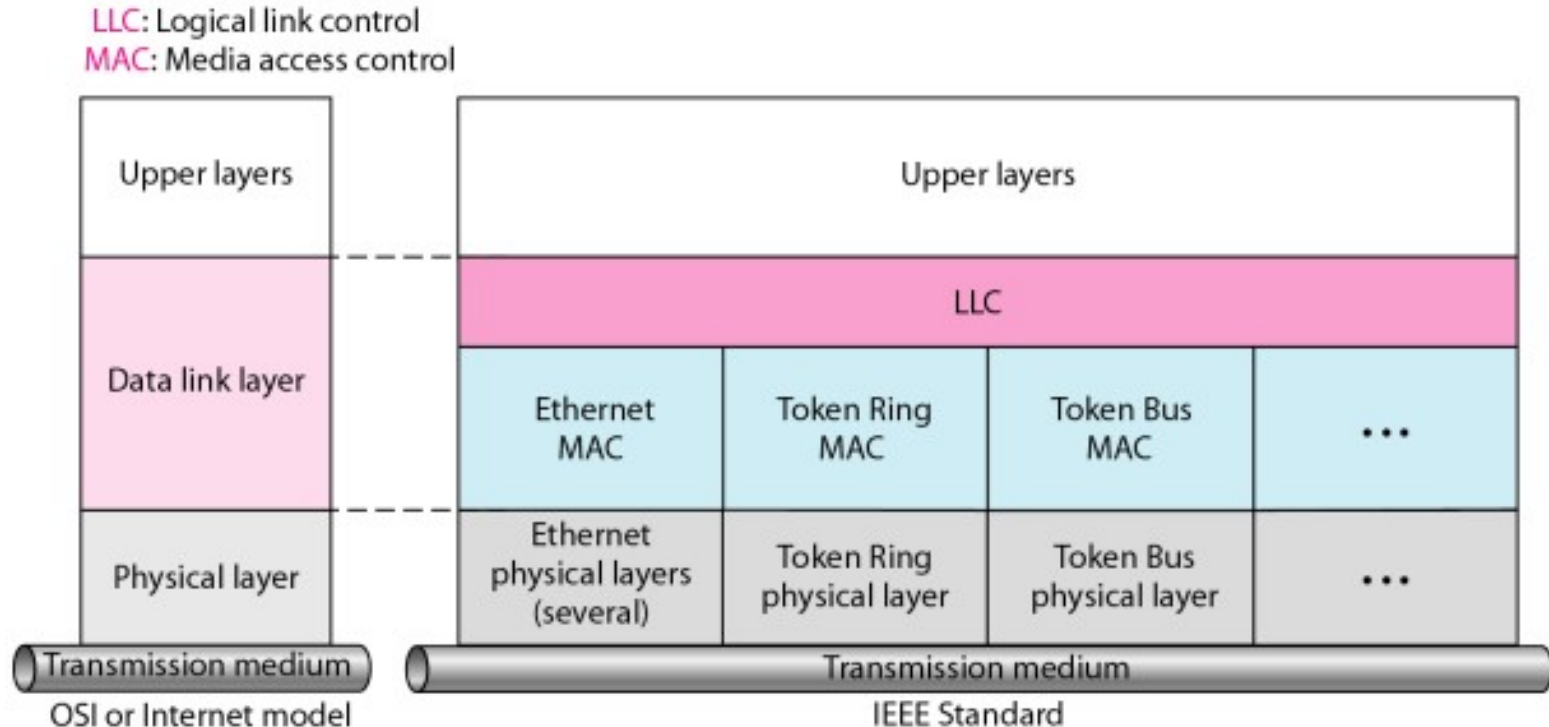
- Recall: TCP/IP protocol suite does not define any protocol for the data link or the physical layer, it accepts any protocol at these layers that can provide services to the network layer
- Data link and physical layer are actually the territory of the local and wide area network
- Several types of LANs were used in the the 1980s and 1990s and used the media access method to solve the problem of sharing the media
 - Ethernet used CSMA/CD
 - Token ring, token bus, FDDI used token passing approach

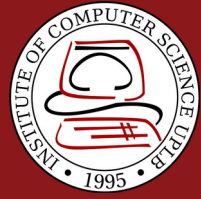
Introduction



- [Project 802](#) – to set the standards to enable intercommunication among equipment from a variety of manufacturers (1985)
- Does not seek to replace OSI but to provide a way of specifying functions of the physical layer and data link layer of major LAN protocols
- Adopted by ANSI and ISO (8802)
- Data link layer divided into [logical link control](#) (LLC) and [media access control](#) (MAC)

IEEE Standards for LANs



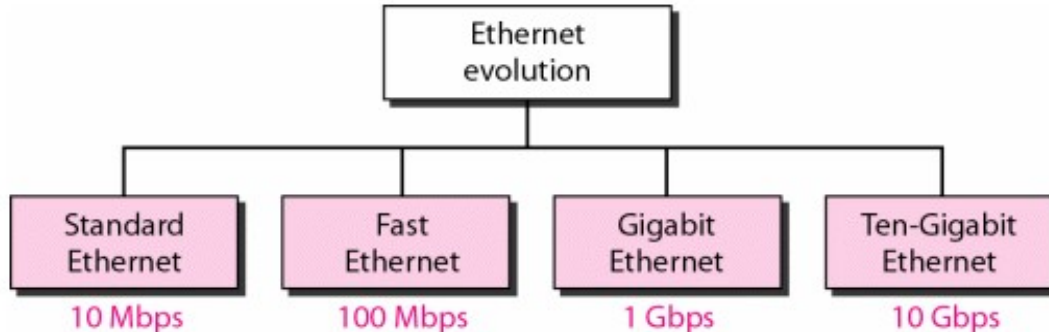


Ethernet Evolution

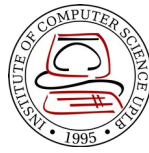
Ethernet



- Original Ethernet was created in 1976 by Robert Metcalfe and David Boggs at Xerox's Palo Alto Research Center (PARC)
 - Standard Ethernet (10 Mbps)
 - Fast Ethernet (100 Mbps)
 - Gigabit Ethernet (1 Gbps)
 - Ten-Gigabit Ethernet (10 Gbps)

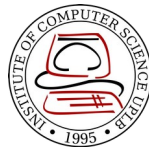


Standard Ethernet



- 802.3
- Connectionless and unreliable service
 - Each frame is independent of the previous or next frame
 - Has no connection establishment or connection termination phases
 - If a frame is corrupted during transmission and the receiver finds about the corruption (CRC-32), the receiver drops the frame silently
 - It is the duty of the high-level protocol to find out about it

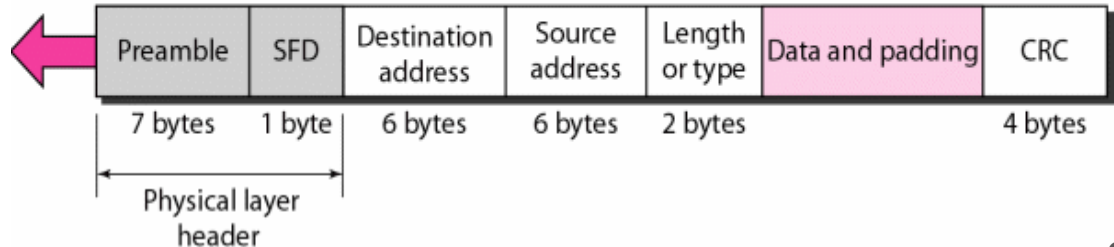
Standard Ethernet



- **Preamble** – 7 bytes (56 bits) of alternating 0s and 1s, for synchronization
- **Start of frame delimiter (SFD)** – 10101011, signals the beginning of the frame
- **Destination address (DA)** – 6 bytes
- **Source address (SA)** – 6 bytes
- **Length** or type – type of upper layer protocol or number of bytes in data field
- **Data** – carries data encapsulated from the upper-layer protocols, minimum of 46 and maximum of 1500 bytes
- **CRC** – CRC-32 for error detection

Preamble: 56 bits of alternating 1s and 0s.

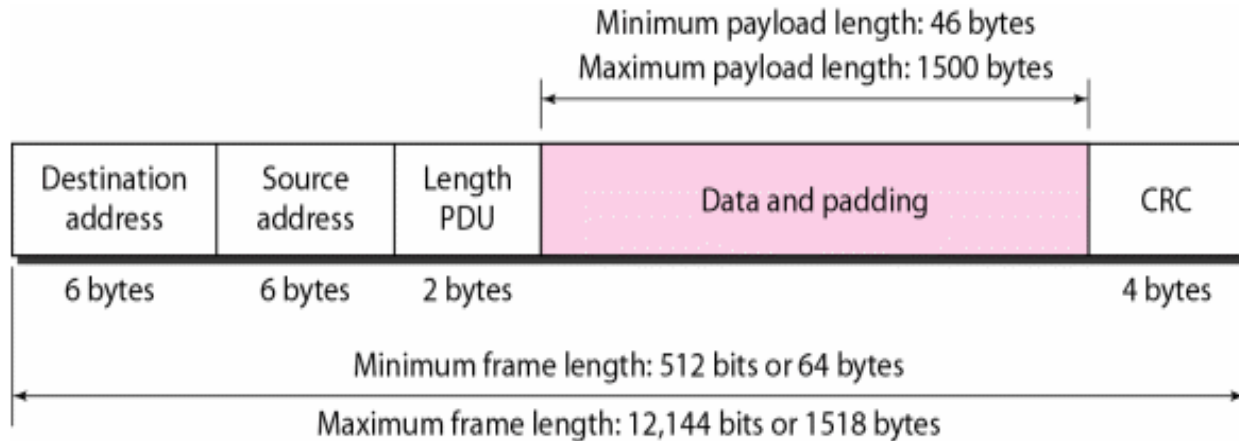
SFD: Start frame delimiter, flag (10101011)



Standard Ethernet



- Has minimum and maximum frame length
- Padding is added if minimum is not met
- Maximum length: reduce buffer size, prevent monopoly of link



Standard Ethernet



- Network Interface Card (NIC) has a 6-byte (48 bits) physical address written in hex with colon between bytes

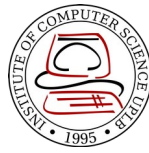
06 : 01 : 02 : 01 : 2C : 4B

└──┘
6 bytes = 12 hex digits = 48 bits

- Destination address can be **unicast**, **multicast**, or **broadcast**; least significant bit of first byte determines the type; broadcast has all 1s



Standard Ethernet

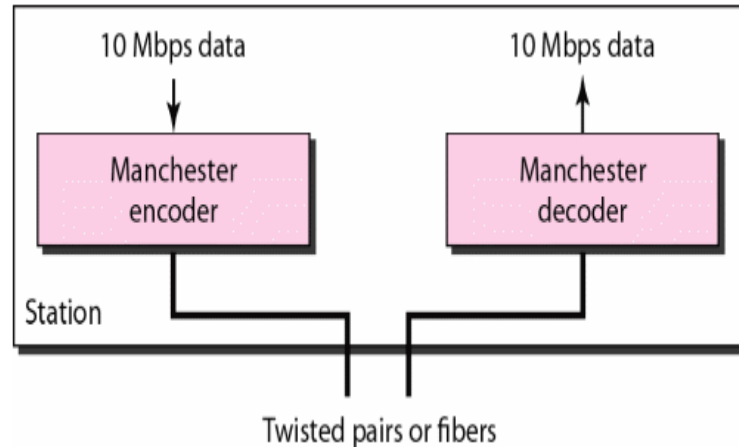


- Examples
 - 4A:30:10:21:10:1A → unicast (A is 1010)
 - 47:20:1B:2E:08:EE → multicast (7 is 0111)
 - FF:FF:FF:FF:FF:FF → broadcast
- Address is sent left to right: 47:20:1B:2E:08:EE
← 11100010 00000100 11011000 .. and so on

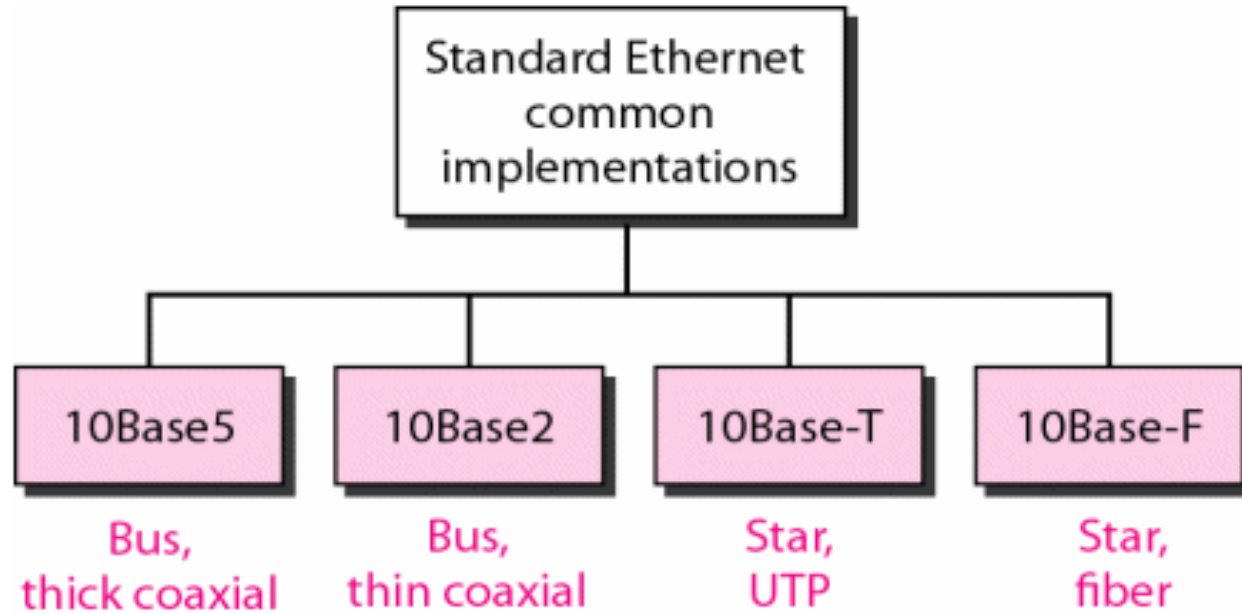
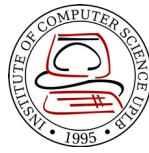
Standard Ethernet



- Uses 1-persistent CSMA/CD
- Physical Layer implementations
 - Uses digital signaling (baseband) at 10Mbps
 - Uses Manchester encoding (self-timing)



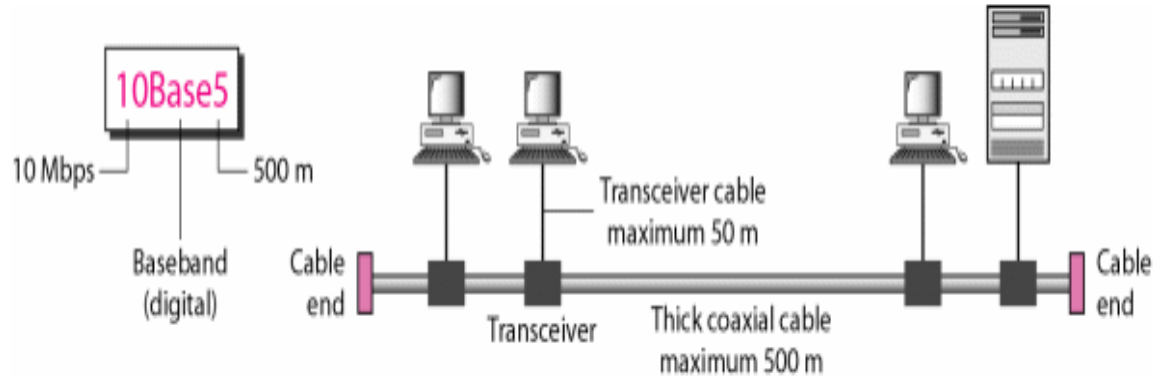
Standard Ethernet



10Base5



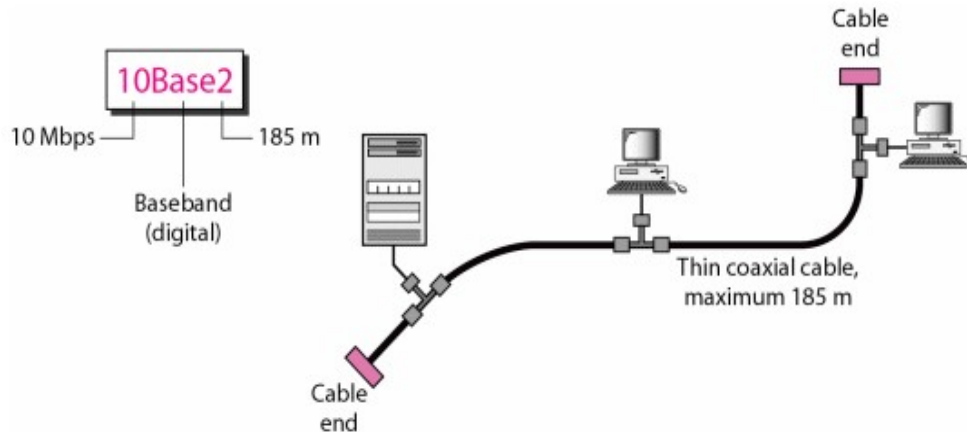
- Thick Ethernet, Thicknet
- Uses bus topology using transceivers (transmitter/receiver)
- Max length of coax must not exceed 500m
- Maximum of five segments connected with repeaters



10Base2



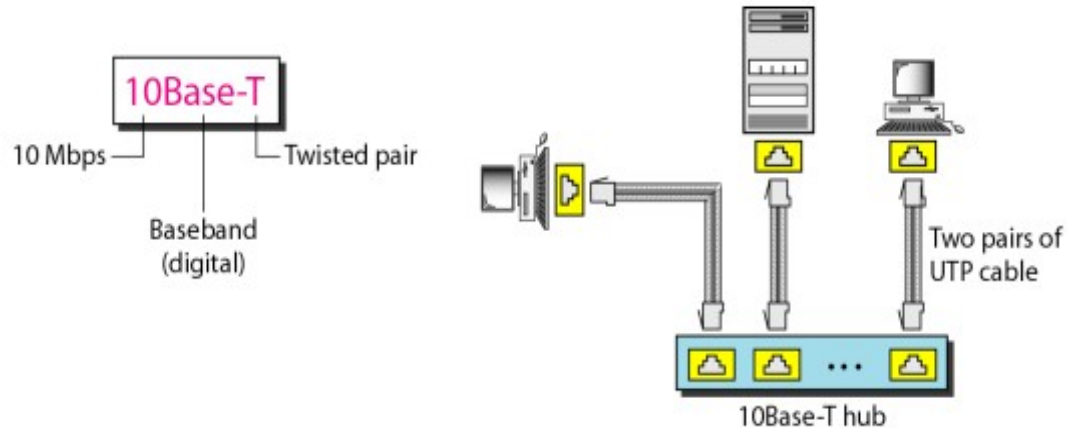
- [802.3a](#); Thin Ethernet, Thinnet, Cheapernet
- Transceiver part of network interface
- Less expensive, easy installation, length cannot exceed 185m
- Collisions happen in the cable



10Base-T



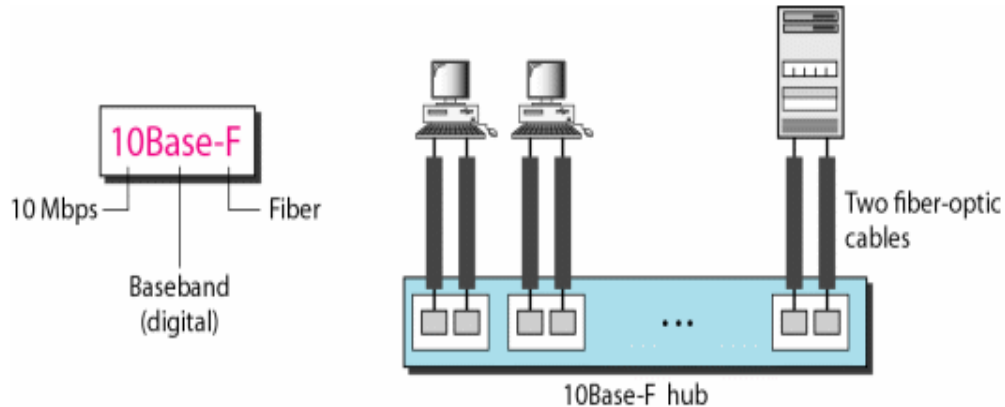
- 802.3i; Twisted-pair Ethernet
- Physical star topology; collisions happen in the hub
- Maximum length is 100m



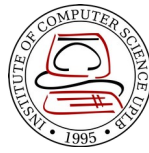
10Base-F



- [802.3j](#); Uses star topology to connect stations to hub
- Max length is 2km
- Used in environments that are electronically noisy



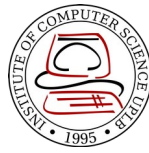
Standard Ethernet



Summary

<i>Implementation</i>	<i>Medium</i>	<i>Medium Length</i>	<i>Encoding</i>
10Base5	Thick coax	500 m	Manchester
10Base2	Thin coax	185 m	Manchester
10Base-T	2 UTP	100 m	Manchester
10Base-F	2 Fiber	2000 m	Manchester

Bridged Ethernet

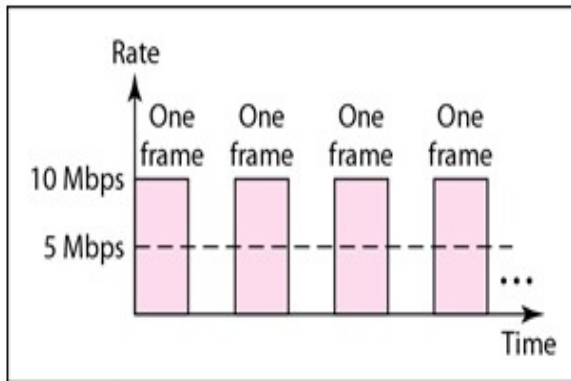


- **Bridge** – network device that connects multiple network segments; uses forwarding database to send frames across network segments
- Two effects of bridges
 - Raise the bandwidth
 - Separate collision domains

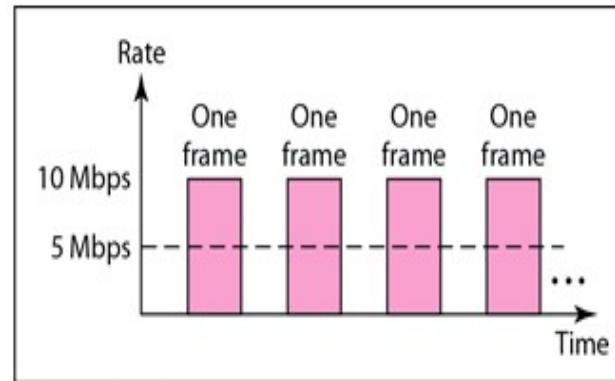
Bridged Ethernet



- In unbridged Ethernet, 10Mbps is shared among all stations with a frame to send
- If two stations has a lot of frames to send, they alternate in usage; on the average each stations sends 5 Mbps
- A bridge divides the network into two or more networks (raising the bandwidth)

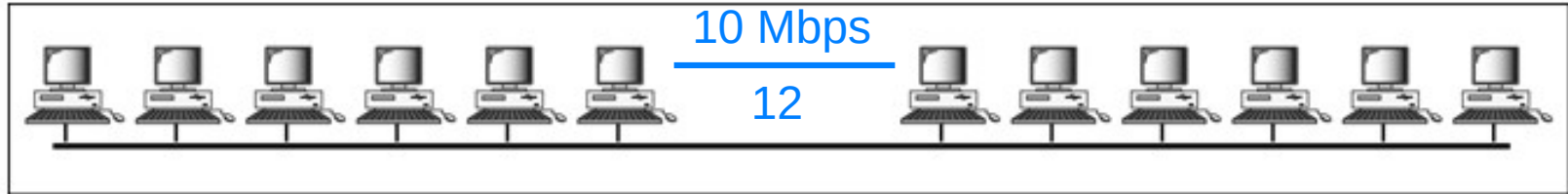


a. First station

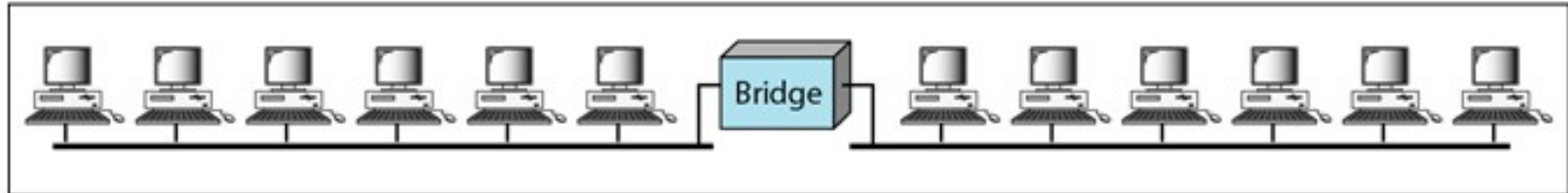


b. Second station

Bridged Ethernet



a. Without bridging



b. With bridging

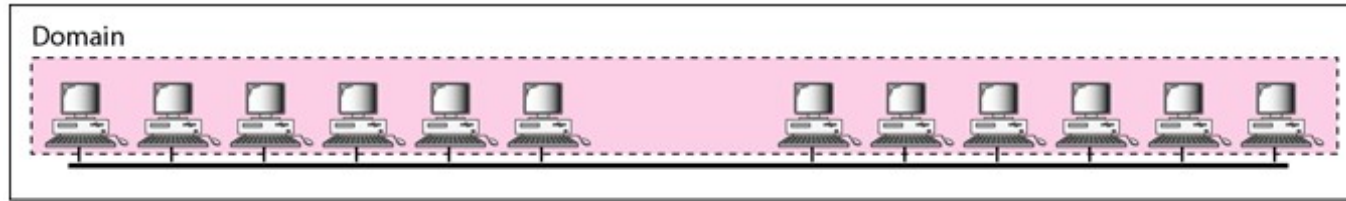
$$\frac{10 \text{ Mbps}}{7}$$

$$\frac{10 \text{ Mbps}}{7}$$

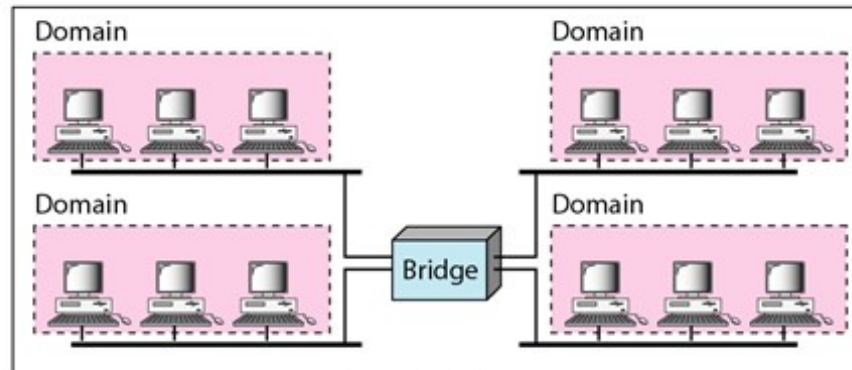
Bridged Ethernet



Collision domains



a. Without bridging

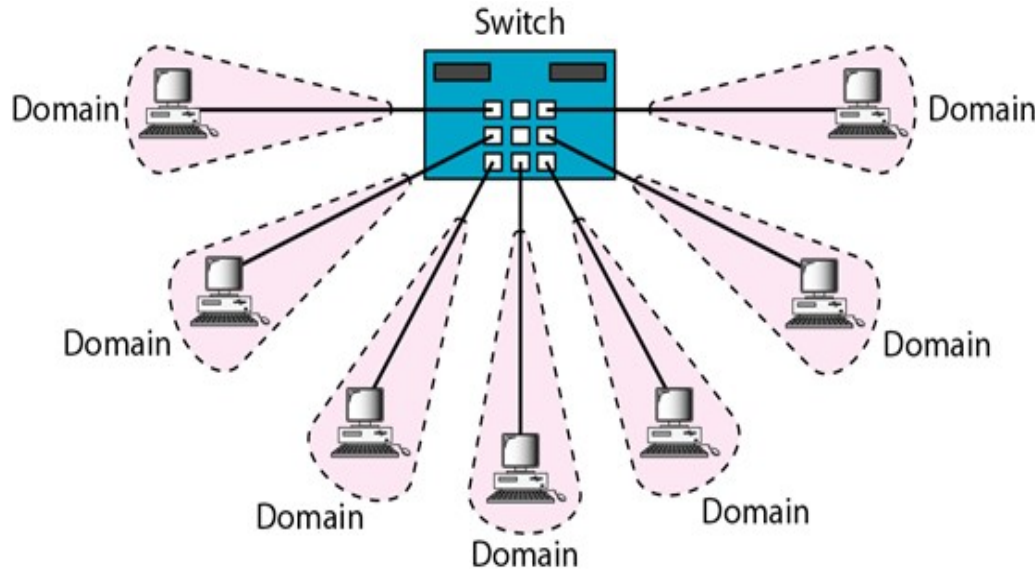


b. With bridging

Switched Ethernet



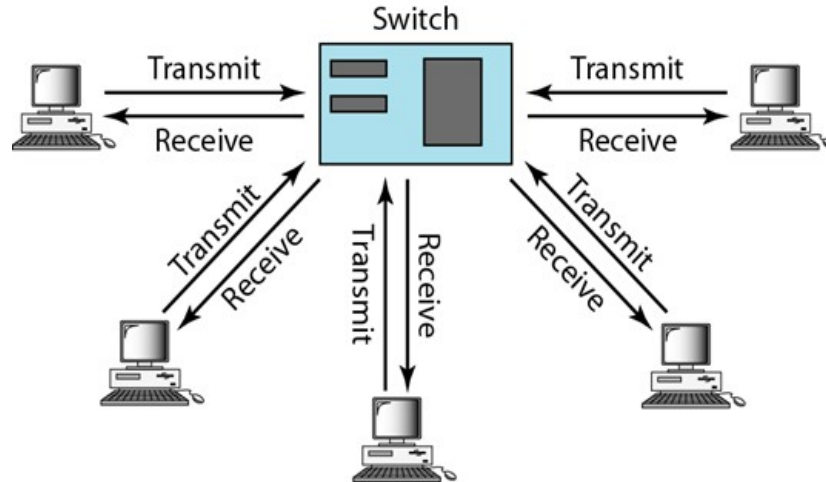
A **layer 2 switch** is an N-port bridge with additional sophistication that allows faster handling of packets; bandwidth is shared only between the station and the switch



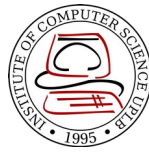
Full Duplex Ethernet



- 10Base5 and 10Base2 are half-duplex (limitation)
- Full duplex mode increases the capacity of each domain from 10 to 20 Mbps
- Configuration uses 2 links: one to transmit, one to receive
- No need for CSMA/CD

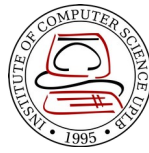


Fast Ethernet



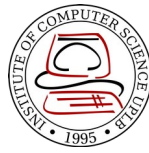
- 802.3u
- Designed to compete with FDDI
- Backwards compatible with Standard Ethernet
- 100 Mbps
- Goals
 - Upgrade the data rate to 100 Mbps
 - Make it compatible with Standard Ethernet
 - Keep the same 48-bit address
 - Keep the same frame format
 - Keep the same minimum and maximum frame lengths

Fast Ethernet



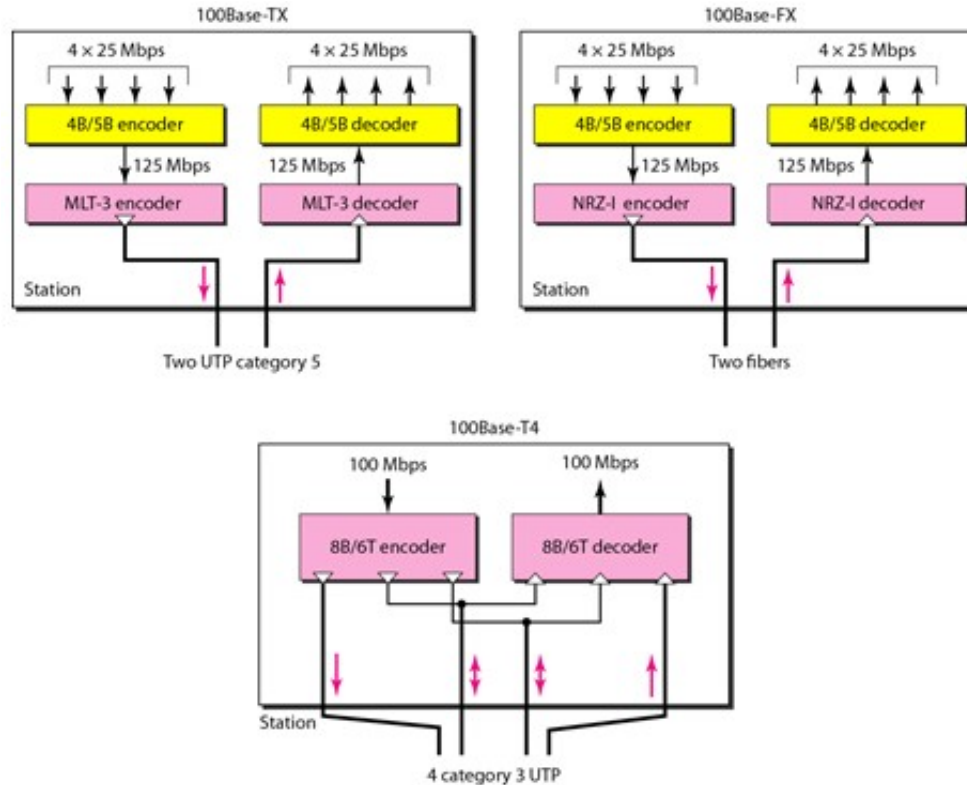
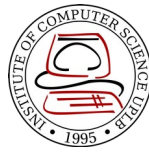
- Keep the MAC sublayer untouched
- Only supports star topology
- CSMA/CD for full-duplex
- **Autonegotiation** – allows two devices to negotiate the mode or data rate of operation
 - Allow incompatible devices to connect (10 Mbps and 100Mbps)
 - Allow device to have multiple capabilities
 - Allow a station to check hub's capabilities

Fast Ethernet

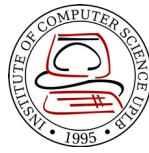


- Topology may be point-to-point if only two stations
- 100Base-TX – two wires CAT 5 UTP
- 100Base-FX – two wires fiber
- 100Base-T4 – four wires CAT 3 UTP
- Manchester encoding is not suitable (needs 200-Mbaud)

Fast Ethernet



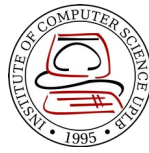
Fast Ethernet



Summary

<i>Implementation</i>	<i>Medium</i>	<i>Medium Length</i>	<i>Wires</i>	<i>Encoding</i>
100Base-TX	UTP or STP	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

Gigabit Ethernet

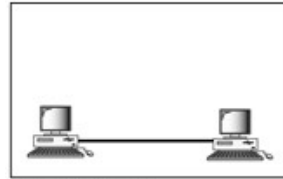


- [802.3z](#); GbE, 1GigE, 1000Mbps
- Goals
 - Upgrade data rate to 1 Gbps
 - Make it compatible with Standard or Fast Ethernet
 - Use the same 48-bit address
 - Use the same frame format
 - Keep same minimum and maximum frame lengths
 - To support autonegotiation as defined in Fast Ethernet

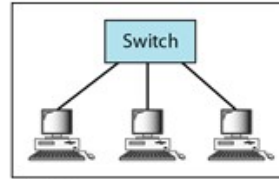
Gigabit Ethernet



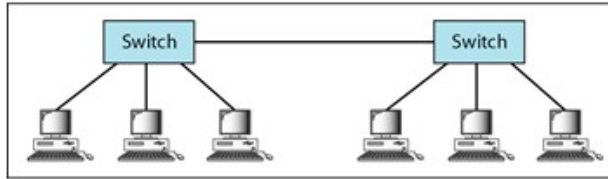
Topology



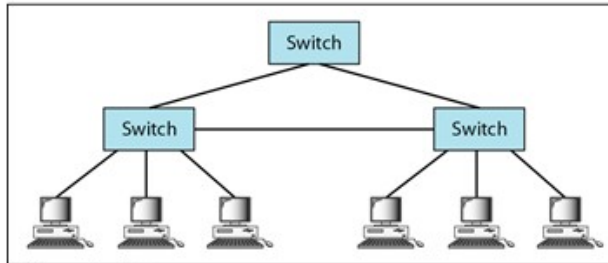
a. Point-to-point



b. Star

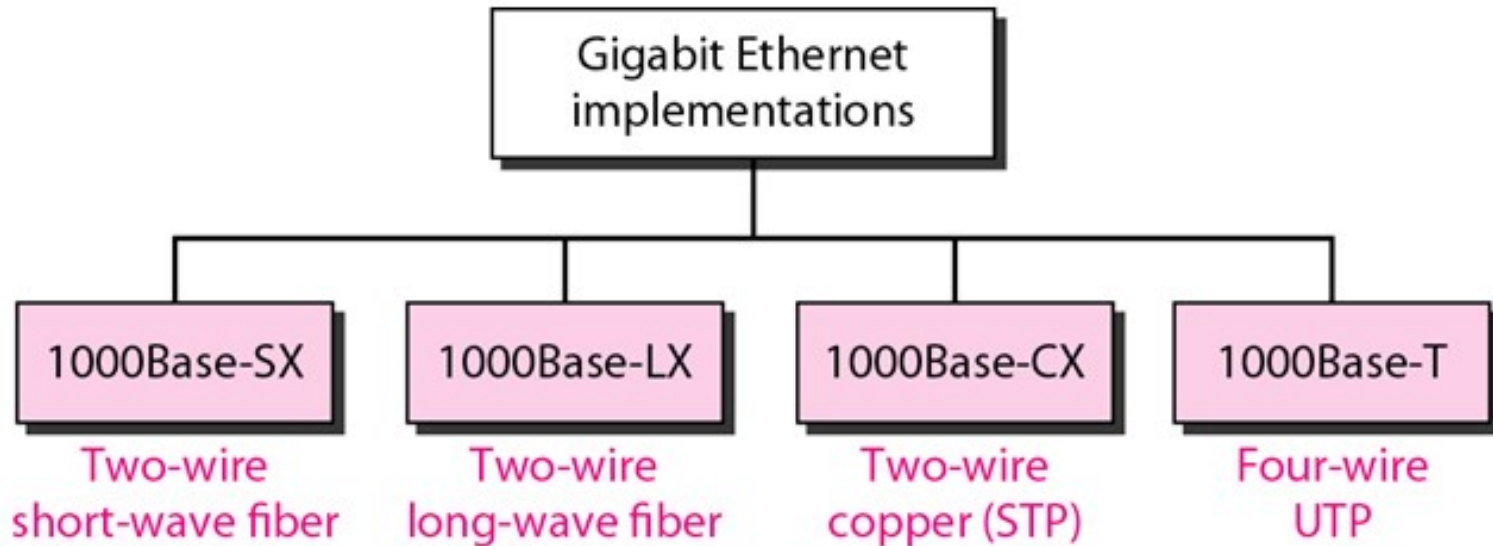


c. Two stars

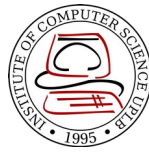


d. Hierarchy of stars

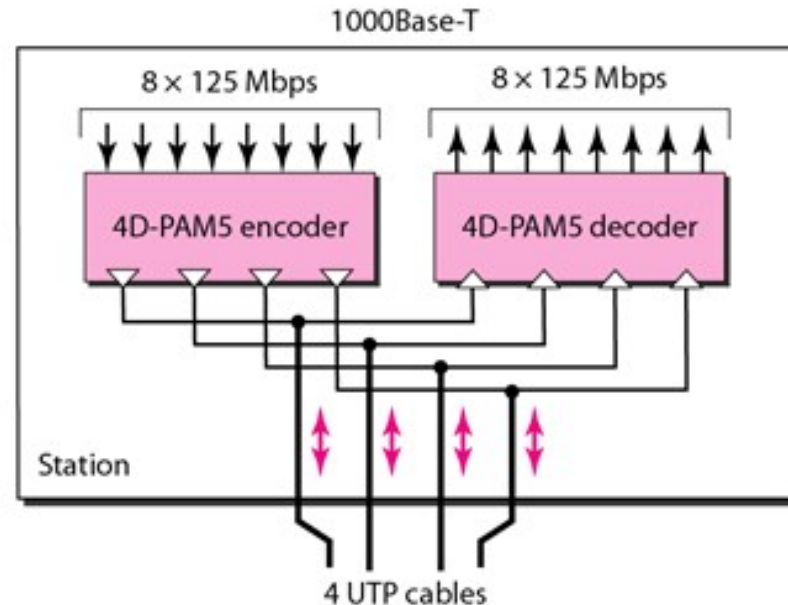
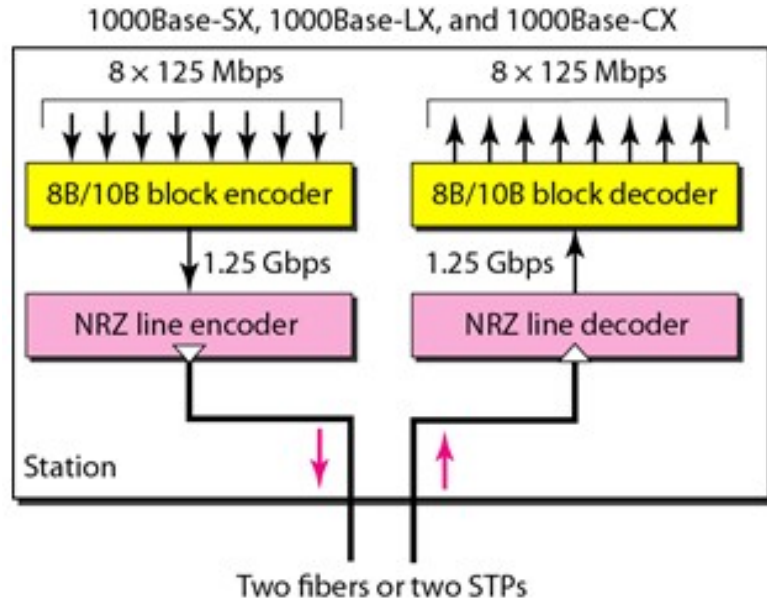
Gigabit Ethernet



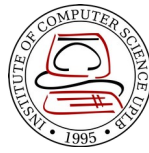
Gigabit Ethernet



Encoding



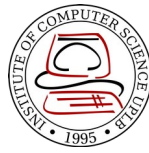
Gigabit Ethernet



Summary

<i>Implementation</i>	<i>Medium</i>	<i>Medium Length</i>	<i>Wires</i>	<i>Encoding</i>
1000Base-SX	Fiber S-W	550 m	2	8B/10B + NRZ
1000Base-LX	Fiber L-W	5000 m	2	8B/10B + NRZ
1000Base-CX	STP	25 m	2	8B/10B + NRZ
1000Base-T4	UTP	100 m	4	4D-PAM5

Ten-Gigabit Ethernet



- 802.3ae; 10GE, 10GbE, 10 GigE
- Goals
 - Upgrade data rate to 10 Gbps
 - Make it compatible with Standard, Fast, and Gigabit Ethernet
 - Use the same 48-bit address
 - Use the same frame format
 - Keep same minimum and maximum frame lengths
 - Interconnect LANs to MANs or WANs
 - Make Ethernet compatible with Frame Relay and ATM

Ten-Gigabit Ethernet



Summary

<i>Implementation</i>	<i>Medium</i>	<i>Medium Length</i>	<i>Number of wires</i>	<i>Encoding</i>
10GBase-SR	Fiber 850 nm	300 m	2	64B66B
10GBase-LR	Fiber 1310 nm	10 Km	2	64B66B
10GBase-EW	Fiber 1350 nm	40 Km	2	SONET
10GBase-X4	Fiber 1310 nm	300 m to 10 Km	2	8B10B

References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.